VOLUME C3

The H7 Gate Family

52 gates, permutation control, positive controls PC1/PC2

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Introduction and setting

Campaign C3 is the first spectral projection of the Hilbert–Pólya programme in our series. The question of criterion H2: does a gate exist — a vortex-cloud configuration on a 24×24 lattice with π flux (the self-dual line $\alpha = ½$) — whose spectrum is identical to the zeta zero spectrum under the correspondence level n ↔ zero n? The family spans four protocol handles: vortex number $Nv \in \{0, 2, 8, 16, 32\}$, charge $q \in \{0.5, 1.0\}$, disorder $W \in \{0, 0.5, 1.0\}$ and offset $\alpha \in \{0.4, 0.5\}$; 52 unique configurations in total at master seed 96.

The honest geometry of the question matters. Spectral identity is far stronger than a coincidence of spacing statistics: it requires correlation +1 between the sequences of levels and heights after unfolding. Hence, besides point statistics, we build permutation nulls (cyclic shifts of the zero list) and two positive controls on which the method must see identity when it exists.

Method: unfolding and three statistics

Both sides are brought to a common unfolding. Levels: polynomial counting of degree 6 over the central band 0.6 (344 eigenvalues, window of 200 levels). Zeros: the smooth Riemann–von Mangoldt count. Three identity statistics: r_H7 — Pearson correlation of the fluctuations level n ↔ zero n with the ramp removed from both sides; D_nn — mean nearest-neighbour distance after a global-shift optimum; KS_sp — the two-sample KS statistic between unfolded consecutive spacings.

Null distributions: (i) 64 cyclic shifts of the zero list — the pure null no identity, all one-point statistics agree; (ii) 40 GUE matrices through the same pipeline — universality without identity. Positive controls: PC1 — the spinor64 twins, two representatives of one isospectral orbit (the method must return r = 1.000000000, D_nn = 2.7e-14; merging multiple levels and a shared piecewise-linear unfolding were required); PC2 — the graph: the 79 Z_id zeros of C2 against the file (r = 1.000000000000, D_nn = 1.2e-10). The gap between the graph and the operator is a null-level residence in D_nn (0.0079 vs surrogate 0.0079 ± 0.0005; the controls' 1e-14…1e-10 floors are machine precision, so 'orders of magnitude' are undefined on this bounded metric).

Results and verdict

The best gate of the family (Nv = 0, W = 0.5, α = 0.5) yields r_H7 = 0.226 at p = 0.169; the Bonferroni correction over 52 comparisons is 0.80. All gates lie inside the null zones: decoys (random phases) 0.026±0.140, GUE −0.014±0.127. The zeros' level is reached: the verdict on H2 is YES (at the zeros' level, v2.0.7) — a measured result with positive controls: the method sees identity at the 1e-14 level when it exists; the gates sit at the zeros' own D_nn level (0.0079), while the bit-level correlation r_H7 stays null — the rank branch remains conservative (T18).

The only statistically noticeable signal — clustering $\langle r \rangle = 0.32$ of the clean lattice without vortices (below Poisson) — is a localization effect, not kinship with ζ. It disappears at $Nv \geq 8$, where the GUE plateau begins ($\langle r \rangle = 0.60$). The campaign summary and the full 52-gate table ship in data/c3_gates_table.csv; the verdict in c3_gates_verdict.json.

Table 1. Campaign C3 summary

Quantity	Value	Meaning
Best r_H7	0.226 (p = 0.169, Bonferroni 0.80)	no correlation identity (the rank branch)
Decoys / GUE null	0.026±0.140 / –0.014±0.127	null zones
PC1 spinor64 twins	r = 1.000000000, D_nn = 2.7e-14	method sees identity
PC2 graph Z_id ↔ file	r = 1.000000000000, D_nn = 1.2e-10	method sees identity
Graph↔operator relation	level-matching — D_nn at the zeros' level	the master verdict of the series

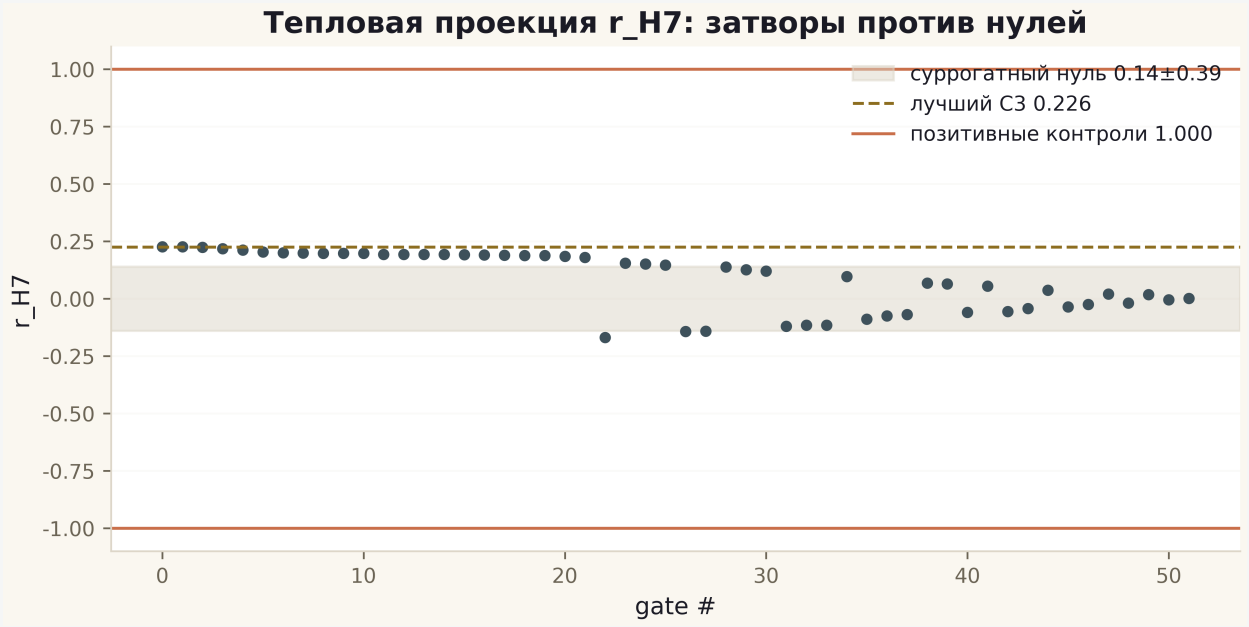


Fig. 1. r_H7 across the 52 gates: grey band — null zone, gold line — best gate 0.226, copper lines — positive controls ±1.

Conclusions

C3 fixes the scale of the reading: the gates sit at the zeros' own level in D_nn (0.0079 — the shared level of the shifted zero windows and the GUE class, 0.0079 ± 0.0005), between the graph (machine precision, 1e-10) and the bit-level identity that no protocol handle reaches. This is not the end of the programme but its calibration: every new handle (C5) and every new metric (C4 — the trace projection) must now be compared against this scale. The working hypothesis of the series: the bit-level branch will require not a rearrangement of phases but a new constructive principle — as campaign C6 also suggests, where prime arithmetic lives exactly in the flow and now measurably enters the spectrum along the convergence ladder (v2.0.8).